

lequ $i_1 i_2 i_3 i_4 \rightarrow$

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\tilde{\mu} y_e^{\text{pr}} y_e^{i_1 i_2} y_u^{i_3 i_4} \overline{a}_e^{\text{pr}} \text{LF}_{3,1,0} [m_l^{\text{p}}, m_e^{\text{r}}] - 3 \tilde{\mu} y_e^{\text{pr}} y_e^{i_1 i_2} y_u^{i_3 i_4} \overline{a}_e^{\text{pr}} \text{LF}_{4,1,-1} [m_l^{\text{p}}, m_e^{\text{r}}] + \right. \\
& 2 \tilde{\mu} y_e^{\text{pr}} y_e^{i_1 i_2} y_u^{i_3 i_4} \overline{a}_e^{\text{pr}} \text{LF}_{5,1,-2} [m_l^{\text{p}}, m_e^{\text{r}}] + 3 \tilde{\mu} y_d^{\text{pr}} y_e^{i_1 i_2} y_u^{i_3 i_4} \overline{a}_d^{\text{pr}} \text{LF}_{3,1,0} [m_q^{\text{p}}, m_d^{\text{r}}] - \\
& 9 \tilde{\mu} y_d^{\text{pr}} y_e^{i_1 i_2} y_u^{i_3 i_4} \overline{a}_d^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_d^{\text{r}}] + 6 \tilde{\mu} y_d^{\text{pr}} y_e^{i_1 i_2} y_u^{i_3 i_4} \overline{a}_d^{\text{pr}} \text{LF}_{5,1,-2} [m_q^{\text{p}}, m_d^{\text{r}}] + \\
& 3 \tilde{\mu} y_e^{i_1 i_2} y_u^{\text{pr}} y_u^{i_3 i_4} \overline{a}_u^{\text{pr}} \text{LF}_{3,1,0} [m_u^{\text{r}}, m_q^{\text{p}}] - 9 \tilde{\mu} y_e^{i_1 i_2} y_u^{\text{pr}} y_u^{i_3 i_4} \overline{a}_u^{\text{pr}} \text{LF}_{4,1,-1} [m_u^{\text{r}}, m_q^{\text{p}}] + \\
& 6 \tilde{\mu} y_e^{i_1 i_2} y_u^{\text{pr}} y_u^{i_3 i_4} \overline{a}_u^{\text{pr}} \text{LF}_{5,1,-2} [m_u^{\text{r}}, m_q^{\text{p}}] + m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{3,1,0} [\tilde{\mu}, m_1] - \\
& 3 m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] + 2 m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] + \\
& 3 m_2 \tilde{\mu} g_2^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{3,1,0} [\tilde{\mu}, m_2] - 9 m_2 \tilde{\mu} g_2^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] + \\
& 6 m_2 \tilde{\mu} g_2^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] - \frac{1}{4} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_1, m_e^{i_2}, m_l^{i_1}] - \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_1, m_l^{i_1}, m_e^{i_2}] - \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \\
& \text{LF}_{2,2,1,-1} [m_1, m_q^{i_3}, m_u^{i_4}] - \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_1, m_u^{i_4}, m_q^{i_3}] - \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_e^{i_2}] + \frac{1}{4} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_l^{i_1}] + \\
& \frac{1}{12} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i_3}] - \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i_4}] - \\
& \frac{3}{4} m_2 \tilde{\mu} g_2^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_l^{i_1}] - \frac{3}{4} m_2 \tilde{\mu} g_2^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i_3}] - \\
& \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_3, m_q^{i_3}, m_u^{i_4}] - \\
& \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,2,1,-1} [m_3, m_u^{i_4}, m_q^{i_3}] + \frac{1}{2} \tilde{\mu} y_d^{i_3 r} y_e^{i_1 i_2} y_u^{i_4} \overline{a}_d^{\text{pr}} \\
& \text{LF}_{2,1,1,0} [m_d^{\text{r}}, m_q^{\text{p}}, \tilde{\mu}] - \frac{1}{2} \tilde{\mu} y_d^{i_3 r} y_e^{i_1 i_2} y_u^{i_4} \overline{a}_d^{\text{pr}} \text{LF}_{3,1,1,-1} [m_d^{\text{r}}, m_q^{\text{p}}, \tilde{\mu}] - \\
& \frac{1}{4} \tilde{\mu} y_d^{i_3 r} y_e^{i_1 i_2} y_u^{i_4} \overline{a}_d^{\text{pr}} \text{LF}_{2,2,1,-1} [m_d^{\text{r}}, \tilde{\mu}, m_q^{\text{p}}] + \frac{1}{2} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \\
& \text{LF}_{2,1,1,0} [m_e^{i_2}, m_1, m_l^{i_1}] - \frac{1}{2} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{3,1,1,-1} [m_e^{i_2}, m_1, m_l^{i_1}] + \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,1,1,0} [m_l^{i_1}, m_1, m_e^{i_2}] - \frac{1}{2} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \\
& \text{LF}_{3,1,1,-1} [m_l^{i_1}, m_1, m_e^{i_2}] + \frac{1}{2} \tilde{\mu} y_d^{i_3 r} y_e^{i_1 i_2} y_u^{i_4} \overline{a}_d^{\text{pr}} \text{LF}_{2,1,1,0} [m_q^{\text{p}}, m_d^{\text{r}}, \tilde{\mu}] - \\
& \frac{1}{2} \tilde{\mu} y_d^{i_3 r} y_e^{i_1 i_2} y_u^{i_4} \overline{a}_d^{\text{pr}} \text{LF}_{3,1,1,-1} [m_q^{\text{p}}, m_d^{\text{r}}, \tilde{\mu}] - \frac{1}{4} \tilde{\mu} y_d^{i_3 r} y_e^{i_1 i_2} y_u^{i_4} \overline{a}_d^{\text{pr}} \\
& \text{LF}_{2,2,1,-1} [m_q^{\text{p}}, \tilde{\mu}, m_d^{\text{r}}] + \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,1,1,0} [m_q^{i_3}, m_1, m_u^{i_4}] - \\
& \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{3,1,1,-1} [m_q^{i_3}, m_1, m_u^{i_4}] + \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \\
& \text{LF}_{2,1,1,0} [m_q^{i_3}, m_3, m_u^{i_4}] - \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{3,1,1,-1} [m_q^{i_3}, m_3, m_u^{i_4}] + \\
& \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,1,1,0} [m_u^{i_4}, m_1, m_q^{i_3}] - \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \\
& \text{LF}_{3,1,1,-1} [m_u^{i_4}, m_1, m_q^{i_3}] + \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{2,1,1,0} [m_u^{i_4}, m_3, m_q^{i_3}] - \\
& \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{3,1,1,-1} [m_u^{i_4}, m_3, m_q^{i_3}] - \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \\
& \text{LF}_{1,1,1,1,0} [m_1, m_e^{i_2}, m_q^{i_3}, \tilde{\mu}] + \frac{2}{3} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{1,1,1,1,0} [m_1, m_e^{i_2}, m_u^{i_4}, \tilde{\mu}] + \\
& \frac{1}{12} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{1,1,1,1,0} [m_1, m_l^{i_1}, m_q^{i_3}, \tilde{\mu}] - \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \\
& \left. \text{LF}_{1,1,1,1,0} [m_1, m_l^{i_1}, m_u^{i_4}, \tilde{\mu}] + \frac{3}{4} m_2 \tilde{\mu} g_2^2 y_e^{i_1 i_2} y_u^{i_3 i_4} \text{LF}_{1,1,1,1,0} [m_2, m_l^{i_1}, m_q^{i_3}, \tilde{\mu}] \right)
\end{aligned}$$